State-of-the-art study plan — Information Reconciliation in QKD

Four study lines October–November 2026

18 October 2026

Guiding question

How can Information Reconciliation (IR) in QKD be performed efficiently while balancing information leakage, communication overhead, decoding complexity and secret-key generation, given the constraints imposed by the physical link?

Structure

The study follows the IR problem from the channel to the hardware in four lines. Each line closes with the gaps it exposes; a short synthesis merges them into research directions.

Line	Focus	Weeks
1	QKD technology and the link properties that shape IR	W1–W2
2	IR metrics and acceptable operating values	W2–W3
3	Protocols and LDPC code choices (type, rate adaptivity, length)	W3–W4
4	LDPC decoders: algorithms and implementation	W5–W6
—	Cross-line synthesis and research directions	W6

1 Line 1 — QKD technology and the link properties that shape IR

Objective. For each deployment type, establish the channel model seen by the reconciliation stage, its operating range, how fast it drifts, and whether the classical channel allows more than one round.

Questions

- Which equivalent channel does IR face (binary symmetric for DV, Gaussian or binary-input AWGN for CV, q-ary for high-dimensional QKD), and is the noise stationary over one frame?
- What QBER (DV) or SNR (CV) range must one reconciliation engine cover?
- On which time scale does the QBER drift (polarisation and phase drift in fibre, pass geometry and scintillation for satellites, turbulence for terrestrial free space)?
- What is the classical round-trip time, and does it allow interactive or incremental (blind) protocols, or does it force a single one-way message?
- How large is the key block available per session (continuous fibre stream versus one satellite pass), and how does the finite-key penalty interact with the IR frame length?

Topics End-to-end post-processing chain (sifting, parameter estimation, IR, verification, privacy amplification) and the raw, sifted, reconciled and secret key hierarchy; DV versus CV QKD; finite-key analysis; frame hierarchy in which privacy amplification may operate on a block larger than the error-correction frame.

Table 1: Properties to characterise per link type.

Link	Channel seen by IR	Drift to characterise	Latency and rounds	Session / block
Terrestrial fibre, DV	Binary symmetric; QBER range per protocol (BB84 decoy, twin-field, entanglement-based)	Slow polarisation and phase drift; compensation strategy	Low latency; interactive and incremental protocols feasible	Continuous stream
Terrestrial fibre, CV	Gaussian; low SNR at long distance; low code rates	SNR and excess-noise fluctuation	One-way reverse reconciliation; multidimensional schemes	Very long frames
LEO satellite	Binary symmetric with time-varying QBER within a pass	Geometry, scintillation, decoy intensities	Moderate latency; on-board computation and downlink bandwidth limit interaction	Short pass; single-pass finite-key regime
GEO satellite	Binary symmetric	As for LEO	High latency; interactive IR likely excluded	Longer visibility
Terrestrial free space (over the air)	Binary symmetric or Gaussian with fading transmittance	Fast turbulence (scintillation, beam wander)	Low latency; adaptive or post-selected reconciliation	Weather-limited sessions

Extraction fields link type; protocol; channel model; QBER or SNR range; drift time scale; QBER estimation method (sampling, syndrome-based, a-priori channel model); classical round-trip time; rounds allowed; session length; finite-key parameters; reconciliation method used.

Expected outcome A per-link requirement profile (QBER or SNR range, drift rate, rounds allowed, block budget) used as the operating envelope by Lines 2–4.

Gaps to record IR designed for stationary channels versus drifting links; satellite and free-space studies that report the QBER but treat reconciliation only briefly; frame length chosen without reference to the finite-key block.

2 Line 2 — IR metrics and acceptable operating values

Objective. Fix one evaluation framework with definitions, the trade-offs between metrics, and acceptable values per link type, so that the work reviewed in Lines 3 and 4 can be compared on equal terms.

Questions

- How are efficiency (f for DV, β for CV) and leakage defined, and what is counted (syndrome, verification hash, interactive messages, bits revealed in blind protocols)?
- Which values are acceptable in practice for each link type, and how do f or β, FER, latency, rounds and throughput trade against each other?
- Which metrics are reported consistently, and which are usually missing?

Definitions to standardise DV: $f = \mathrm{leak}_{\mathrm{IR}}/(n\,h(Q))$, with $f = 1$ at the Slepian–Wolf limit. CV: $\beta = R/C$ with $C = \frac{1}{2}\log_2(1+\mathrm{SNR})$. The effective key rate per frame scales with $(1-\mathrm{FER})$; leakage includes every public message, including those of two-way protocols. FER is reported with confidence intervals.

Table 2: Metrics and the values to collect.

Metric	Values to collect	Notes
f (DV)	Achieved f versus QBER; tolerated f in deployed systems	Report at the operating QBER, not only at the code's design point
β (CV)	Achieved β versus SNR	CV codes are often run at high FER to raise β
FER	Operating FER per link type; whether failures are detected and discarded	Joint (f or β, FER) optimum rather than FER alone
Rounds	One-way, blind (incremental), interactive	Latency $\approx$ rounds $\times$ round-trip time plus decoding time
Latency	Per-frame decoding latency; post-pass processing window for satellites	Rarely reported
Throughput	Decoder throughput relative to the sifted-key rate	Decoder versus system throughput
Energy	Throughput per watt	Seldom reported
QBER estimation cost	Fraction of key spent on parameter estimation	Competes with IR leakage

Extraction fields definition of f or β used; leakage terms counted; FER and number of frames simulated; rounds; latency; throughput and platform; energy; QBER or SNR at which each value holds.

Expected outcome A metric sheet with definitions and acceptable values per link type (Table 2 completed), used as the comparison template in Lines 3–4.

Gaps to record inconsistent definitions of f (with or without the hash or interactive bits); FER reported without confidence intervals; latency and energy missing; efficiency reported at a single QBER only.

3 Line 3 — Protocols and LDPC code choices

Objective. Understand how protocols and code choices realise the operating envelope of Line 1 under the metrics of Line 2, with emphasis on LDPC code type, rate adaptivity and code length.

Questions

- Why did the field move from interactive methods (Cascade, Winnow) to one-way codes, and where is interaction still preferable?
- Which rate-adaptation scheme fits which link: a pool of fixed-rate matrices, puncturing and shortening of a mother code, blind (incremental) reconciliation, rateless protograph codes, row combining, or selection driven by channel estimates?
- Which code family per regime: high-rate QC or protograph LDPC for DV; multi-edge-type and Raptor-like LDPC for low-rate CV; non-binary LDPC for high-dimensional QKD; spatially coupled LDPC; polar codes as the main alternative?
- How does code length trade efficiency against latency and memory, from short frames based on standard codes to very long frames for spatially coupled and CV codes?
- How are degree profiles and puncturing patterns designed (density evolution, rate-adaptive protograph design, progressive edge growth lifting, finite-length optimisation of coupled codes)?

Expected outcome A comparative table of protocols and codes, indexed by the link types of Line 1, with the efficiency, FER and rounds each achieves.

Table 3: Code-choice dimensions to tabulate for every paper.

Dimension	Values to record
Protocol class	interactive (Cascade, Winnow) / one-way / blind (incremental) / rateless / hybrid (one-way plus interactive clean-up)
Code type	regular or irregular QC-LDPC; protograph; multi-edge type; Raptor-like; spatially coupled; non-binary; polar (reference)
Adaptivity	fixed rate; matrix pool; puncturing/shortening of one mother; incremental syndrome; rateless extension; rate range covered
Length	n, lifting size Z, section size (coupled codes), frames per privacy-amplification block
Design method	density-evolution or EXIT threshold; girth; trapping-set screening; reuse of standard codes (802.11n, DVB-S2, 5G NR)
Results	f or β versus QBER or SNR, FER, rounds, leakage terms counted

Gaps to record rate-adaptive schemes designed for stationary QBER; puncturing and shortening patterns chosen at random rather than designed; high-rate DV codes compared against CV results; spatially coupled LDPC rarely studied for DV-QKD at practical lengths.

4 Line 4 — LDPC decoders: algorithms and implementation

Objective. Understand which decoders deliver the efficiency of Line 3 at the throughput, latency and energy required by Line 1, and which implementation choices limit performance.

Questions

- Which decoding algorithm and schedule: sum-product, min-sum and its normalised and offset forms, self-corrected min-sum, finite-alphabet decoders; flooding versus layered; windowed decoding for coupled codes?
- How much does quantisation cost near the operating QBER, and which low-bit-width methods recover it?
- How many iterations are needed, which stopping rules are used, and what post-processing handles residual failures?
- Which platform reaches the required throughput: CPU with SIMD, GPU, FPGA or ASIC?
- How do memory organisation and pipelining constrain the code, and how does rate adaptivity complicate the datapath?

Extraction fields algorithm; schedule; message and channel bit widths; iterations (maximum and average); early termination; post-processing; platform and clock; throughput (decoded and information); latency; resources or area; power and energy per bit; f or β and FER at the same operating point.

Expected outcome A decoder-implementation table aligned with Table 2, showing the efficiency, throughput and energy of each design.

Gaps to record decoders evaluated at classical operating points rather than at the QKD operating point; throughput reported without f at the same setting; few rate-adaptive hardware decoders; windowed decoding of coupled codes not yet characterised for high-rate DV codes.

5 Cross-line synthesis and research directions

In week 6 the gaps recorded in Lines 1–4 are merged and ranked by evidence. Candidate directions to test against the collected evidence:

1. Channel-aware IR for drifting links (satellite pass, free space): use an a-priori or estimated QBER trajectory to select rate and decoder settings per frame.
2. One-way, high-rate DV reconciliation with efficiency close to the Slepian–Wolf limit at frame lengths and decoders compatible with hardware throughput.
3. Latency-bounded decoding (windowed, layered, finite-alphabet) characterised at the QKD operating point.
4. Designed rather than random rate adaptation: puncturing and shortening patterns and blind increments optimised jointly with the code.
5. A common reporting standard: leakage terms, FER with confidence intervals, latency, throughput and energy at the same operating point.

6 Schedule

Week	Work	Deliverable
W1 (19–25 Oct)	Line 1: post-processing chain, fibre DV and CV, finite-key basics	Table 1, fibre rows
W2 (26 Oct–1 Nov)	Line 1: satellite and terrestrial free space; Line 2: definitions	Table 1 complete; metric definitions
W3 (2–8 Nov)	Line 2: acceptable values; Line 3: interactive, one-way, blind, rateless	Table 2 complete; protocol table
W4 (9–15 Nov)	Line 3: code types, rate adaptivity, code length, design methods	Table 3 filled for all papers
W5 (16–22 Nov)	Line 4: algorithms, schedules, quantisation, post-processing	Algorithm table
W6 (23–29 Nov)	Line 4: platforms and implementation; cross-line synthesis	Implementation table; ranked research directions
30 Nov	Final review document	State-of-the-art report and literature database

Literature database

One record per paper with: bibliographic key; line(s); link type; protocol; channel model and QBER or SNR range; code family, rate(s), length, adaptivity; decoder, bit widths, iterations, platform; f or β, FER (and number of frames), rounds, latency, throughput, energy; leakage terms counted; limitations stated by the authors; relevance to the guiding question.